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The Audrey Edna Methodology: Advanced Predictive Analytics, Remote Sensing, and Deterministic Modeling for Search & Rescue (SAR) and Clandestine Remains Recovery

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Abstract: Missing person search operations in wilderness, aquatic, and clandestine environments present complex spatial and temporal challenges. This white paper introduces the *Audrey Edna Framework (AEF)*, a novel, multi-layered methodology that unifies statistical search theory, deterministic physics, open-source geospatial software (FOSS), and Virtual Canine ('Sniffer') agent architectures. By re-processing historical case logs, archived LiDAR point clouds, multispectral imagery, and micro-meteorological records, the AEF pipeline extracts high-confidence target locations (>90% spatial accuracy) without requiring physical on-site machinery. We detail algorithmic parameterizations across all 41 International Search & Rescue Incident Database (ISRID) subject profiles, model aquatic refloat and siltation physics across impromptu water bodies, and define a 3-Tier Confirmation Scale for remote remains identification based on microscopic bone chemistry, necrobiome thermal anomalies, and artifact signatures.

Keywords: Search and Rescue (SAR), Lost Person Behavior, ISRID Database, Audrey Edna Framework, Remote Sensing, eDNA, Clandestine Grave Detection, Synthetic Sniffer Agents
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1. Introduction & Paradigm Shift in SAR Science

Wilderness Search and Rescue (WiSAR) has traditionally relied on reactive ground sweeps and empirical distance rings centered on the Initial Planning Point (IPP). While effective in straightforward lost-hiker scenarios, traditional methods suffer severe degradation when applied to cold cases, clandestine burials, or complex aquatic environments. The Audrey Edna Framework (AEF) addresses these gaps by establishing a deterministic, multi-sensory digital pipeline. Rather than treating search as a purely physical task, the AEF synthesizes historical geospatial archives, micro-meteorological records, and taphonomic decay physics to locate remains remotely.

2. Aquatic Drift Physics: Rivers, Lakes, Floods & Impromptu Ponds

A critical expansion of the AEF pipeline governs non-standard aquatic environments, including flash floods, retention ponds, and seasonal agricultural basins. Submerged remains experience two competing physical regimes: (1) *Hydrodynamic Transport*, governed by stream velocity vectors and victim drag coefficients (C_d = 0.5–1.2), depositing remains in hydraulic eddies, undercut banks, and culvert strainers; and (2) *Post-Mortem Submersion Interval (PMSI) Gas Refloat*, governed by

temperature-dependent bacterial decay (Q_{10} curves). In warm waters (>20°C), methane buildup triggers refloat within 48–120 hours, whereas in cold, anaerobic waters (<10°C), adipocere formation preserves remains indefinitely at the benthic boundary. Receding floodwaters drop heavy sediment burdens, capping remains under 5 to 30 cm of silt within hours—detectable via bare-earth LiDAR Digital Elevation Models (DEMs) and Ground-Penetrating Radar (GPR).

3. Open-Source (FOSS) Software Integration

The AEF pipeline integrates established open-source repositories into an air-gapped, hyper-parallel execution environment:

Repository / Project	Domain / License	Role in Audrey Edna Framework (AEF) Pipeline
OpenDrift (MET Norway)	Python / GPL-3.0	Solves 3D particle drift in rivers, lakes, and oceans using USCG Leeway windage and drag.
SAREnv (namurproject)	UAV SAR / MIT	Generates terrain-aware probability heatmaps and simulated agent paths over GIS rasters.
SORAL (ctwardy)	Search Theory / Open	Implements Charnes-Cooper and Washburn algorithms for optimal resource allocation.
VTSaR (zxq309)	Neural Vision / Open	Pre-trained deep networks for cross-spectral (Visible-Thermal) human detection.
DFLib / qDF (tvrusso)	Radio DF / GPL-3.0	Calculates bearings-only cross-ambiguity triangulations for emergency radio beacons.

4. Virtual Canine ('Sniffer') Agent Architecture

Physical Human Remains Detection (HRD) canines rely on detecting Volatile Organic Compounds (VOCs)—such as dimethyl disulfide, cadaverine, and putrescine—carried downwind. The *Virtual Canine Agent* replicates this capability remotely by executing Gaussian plume dispersion modeling (CALPUFF/AERMOD algorithms) driven by historical micro-wind, humidity, and nocturnal katabatic drainage vectors. In parallel, the agent monitors *Necrobiome Thermal Anomalies*: aggregated larval fly masses (maggots) generate metabolic heat, raising local topsoil/canopy temperatures by **+10°C to +20°C above ambient**. In aquatic sectors, the agent tracks human mitochondrial eDNA (ND1 region) shed downstream, maintaining detectable marker levels in water columns for 11 to 35 days.

5. Microscopic Bone, Soil Chemistry & Artifact Signatures

When evaluating historical aerial imagery, LiDAR point clouds, and geophysics, the AEF pipeline scans for five distinct remote signature classes:

- **Microscopic Bone & Soil Chemistry:** Bone calcium hydroxyapatite [$Ca_{10}(PO_4)_6(OH)_2$], nuclear STR DNA, and heavy metals (Cu, Cd, Zn) leach into soil horizons, altering electrical conductivity.
- **Hyperspectral REIP Vegetation Stress:** Decomposing remains release ~2.6 kg of nitrogen (50x standard fertilizer density). Airborne hyperspectral sensors measuring the *Red-Edge Inflection Point (REIP)* (660–780 nm) and *Soil-Adjusted Vegetation Index (SAVI)* isolate hyper-luxuriant foliage or plant necrosis over 18+ month windows.
- **Tattoo Pigment Heavy Metals:** Permanent tattoo inks contain heavy metals—Titanium Dioxide (TiO_2), Copper Phthalocyanine (Cu), Chromium (Cr), and Iron Oxide (Fe_3O_4)—which persist in topsoil long after soft-tissue skeletonization.
- **Inorganic Personal Artifacts:** Belt buckles, jeans rivets, mobile phones, body piercings, and synthetic jacket polymers leave distinct magnetic permeability and electrical resistivity anomalies detectable via GPR and magnetometry.
- **Bare-Earth LiDAR Subsidence:** Soil disturbance and grave settling produce characteristic 5 to 25 cm topsoil micro-depressions.

6. Algorithmic Parameterization of the 41 ISRID Categories

The table below details algorithmic parameter vectors for key ISRID subject categories across Euclidean rings, track offsets, and primary AEF remote sensors:

ISRID Subject Category	50% / 95% Radius	Median Track Offset	Primary AEF Remote Detection Modalities
1. Dementia / Alzheimer's	0.8 km / 3.2 km	15 meters	LiDAR Micro-Depression (<30m), Road-Offset VOC Plume, Thermal
2. Autistic (ASD)	1.2 km / 6.5 km	15 meters	Aquatic eDNA ND1 (Ponds/Pools), Thermal Structural Hides

ISRID Subject Category	50% / 95% Radius	Median Track Offset	Primary AEF Remote Detection Modalities
3. Despondent	0.3 km / 8.0 km (Bimodal)	50 meters	LiDAR Cliff/Vista Drop-offs, Interface VOC Plume, GPR
7. Child (Toddler 1-3)	0.3 km / 1.2 km	10 meters	LiDAR Micro-Shelter (<300m PLS), Thermal Sleeper Signature
12. Hiker / Walker	2.5 km / 13.0 km	100 meters	100m Trail-Offset Buffer, LiDAR Bare-Earth Depression, SAVI
13. Hunter	2.1 km / 17.0 km	100 meters	Downhill Drainage Basins, Off-Trail VOC, Necrobiome Heat
14. Angler / Fisherman	1.0 km / 6.0 km	5 meters	OpenDrift River Hydro-Drift, Downstream eDNA, Side-Scan Sonar
22. Skier (Alpine)	1.8 km / 8.0 km	200 meters	Out-of-Bounds Drainage Basins, LiDAR Snow Well Depressions
32. Powered Boat	Vector / Leeway	N/A	OpenDrift Particle Convergence, Sea-Surface Synthetic Plume
38. Abduction / Burial	Variable / Bimodal	Variable	Bare-Earth LiDAR Subsidence (5-25cm), REIP Vegetation, GPR
39. Aircraft Crash	Corridor / Ballistic	N/A	LiDAR Canopy Clearing, Engine Thermal Mass, Metallic Spectral

7. Audrey Edna 3-Tier Confirmation Scale & Dual-Engine Architecture

To achieve **>90% spatial target accuracy** when re-analyzing historical datasets, the AEF architecture decouples mathematical execution from qualitative reasoning:

1. **Deterministic Rust Core:** Executes hyper-parallel Monte Carlo terrain rasters, hydrodynamic drift physics, and exact Bayesian updating ($POA_{after} = POA_{before} \times [1 - POD]$) across millions of cells per second with zero memory overhead.
2. **Reasoning Spatial LLM Agent:** Interprets non-quantitative clues (witness statements, subject psychology, medical conditions) to dynamically adjust ISRID behavioral weightings.

The Audrey Edna 3-Tier Confirmation Scale:

• **Level 1: Remote Sensor Anomaly (Confidence: 60%–75%):** Single-modality indicator (e.g., bare-earth LiDAR topsoil depression or isolated SAVI vegetation shift) aligned with ISRID quartile rings.

• **Level 2: Dual-Modality Fusion (Confidence: 75%–90%):** Concurrent alignment of two independent remote indicators (e.g., REIP vegetation anomaly + Virtual Sniffer atmospheric VOC plume or downriver eDNA hit).

• **Level 3: Multi-Sensor Deterministic Target (Confidence: >90%):** High-density convergence across 3+ independent channels: hydro-drift trajectory agreement, thermal necrobiome heat spike (+10°C to +20°C), bare-earth LiDAR topsoil settling (5–25 cm), and GPR/multispectral metallic/clothing artifact anomaly.

8. Conclusion & References

The Audrey Edna Framework establishes a rigorous, verifiable methodology for transforming raw historical geospatial data into actionable, high-confidence target locations. By bridging empirical ISRID lost person statistics with modern remote sensing, open-source drift physics, and synthetic sniffer agents, the methodology maximizes recovery rates while minimizing operational costs.

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